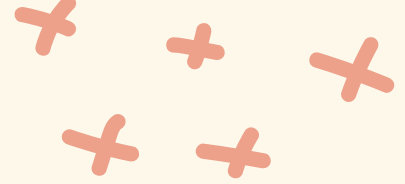
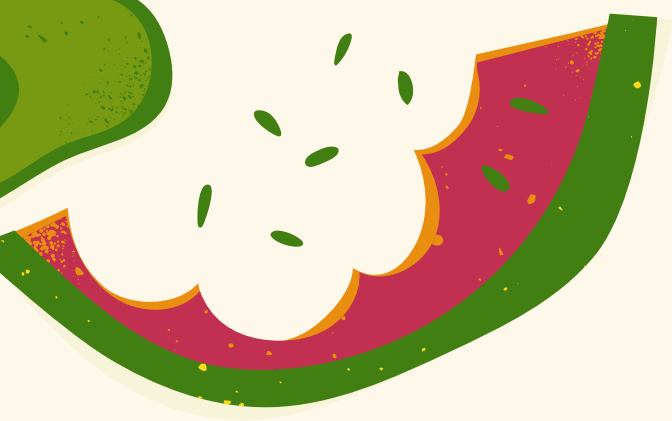


人工智慧應用實驗

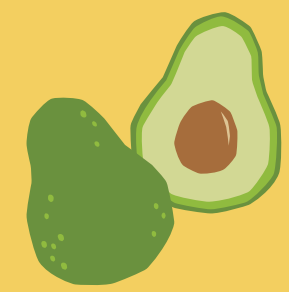
REFRIGERATOR CONSULTANT

A tool can help you solve your
disaster from refrigerator.

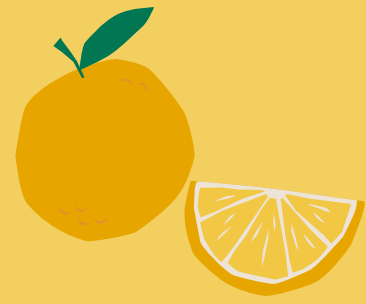




Division of work



411770349 黃妤涵
concept、program、ppt



411770521 繆語
concept、program、ppt



411770547 李佩臻
concept、program、ppt



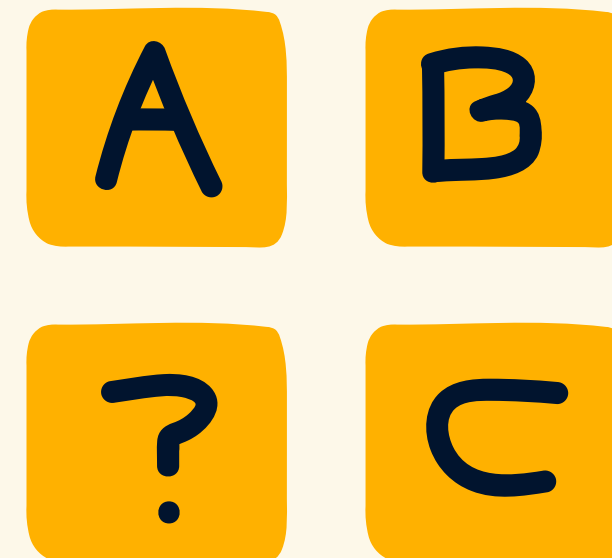
CONCEPT

解決日常煮飯煩惱



PICTURES

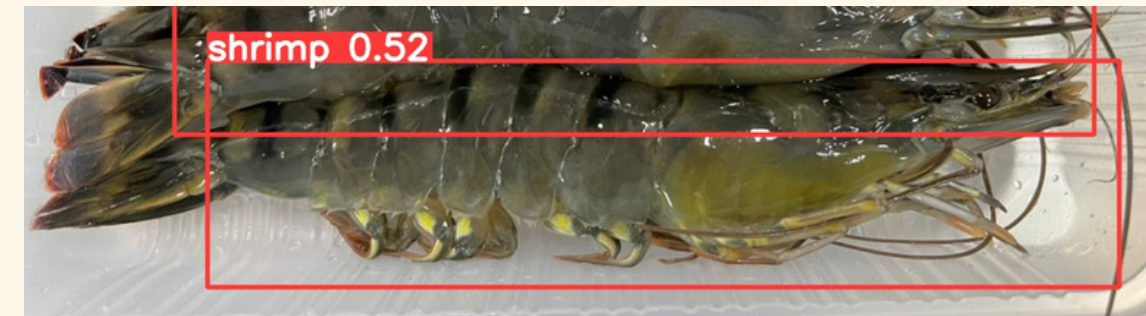
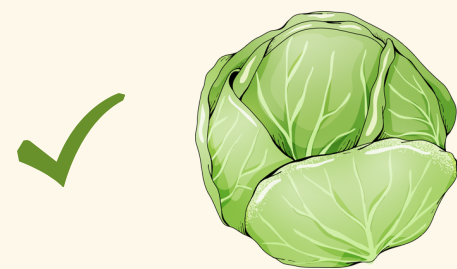
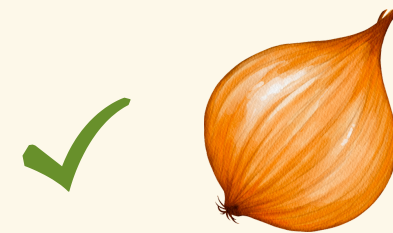
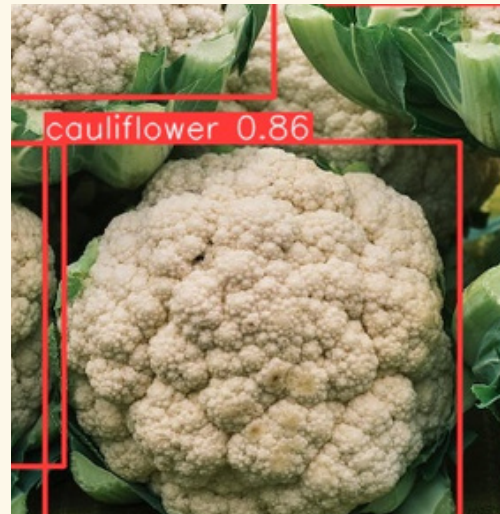
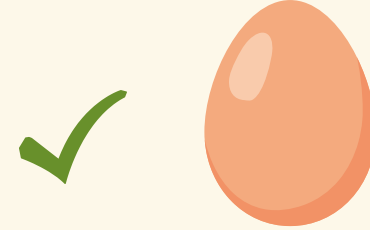
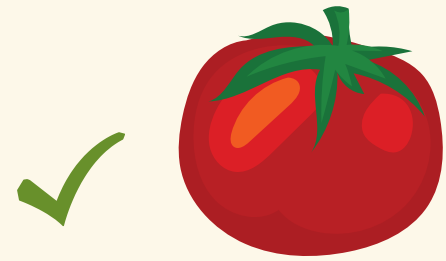
provide a picture contain ingredients



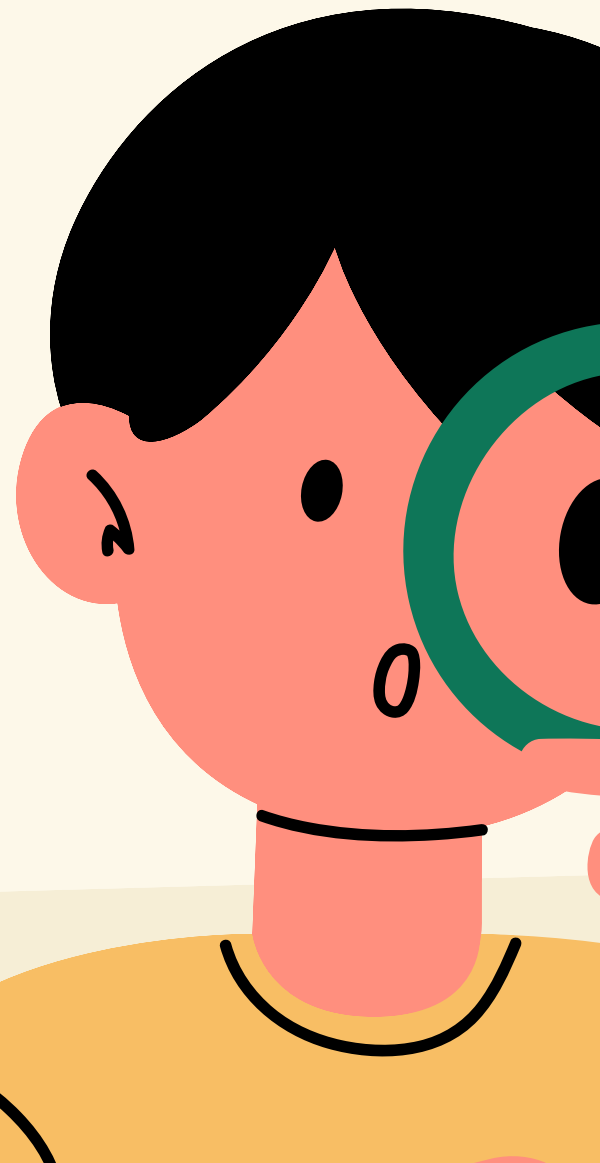
WORDS

enter the words of your ingredients

類別種類：



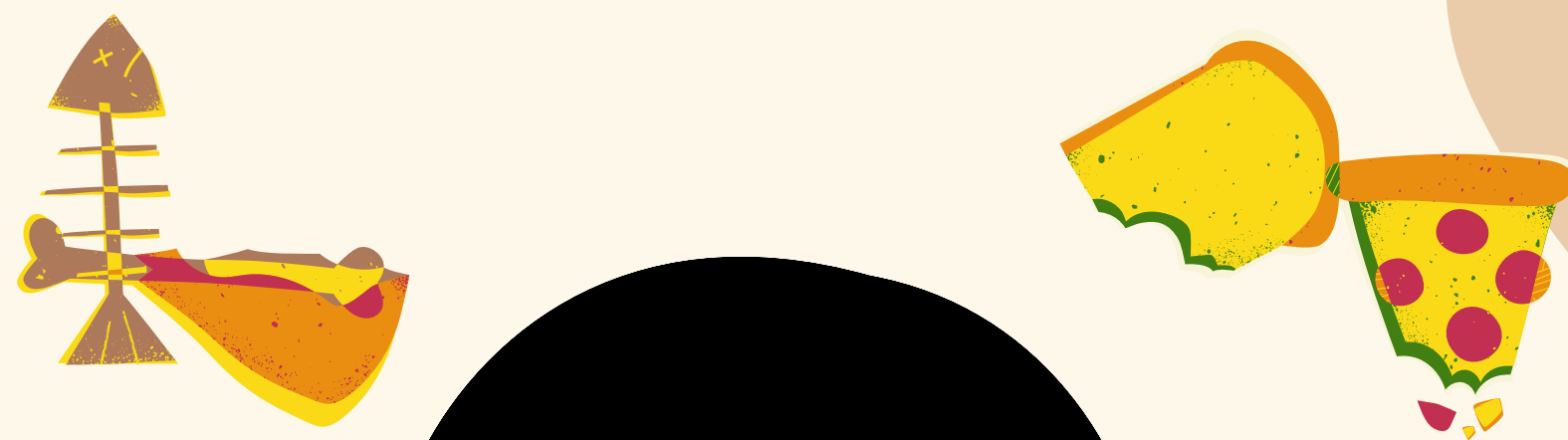
共六種類別
每種約500筆資料進行訓練



實作設計概念：

- ✓ 使用者提供照片或是文字
- ✓ 利用我們訓練的 model 判斷使用者給的圖片中所含的類別
- ✓ 將偵測到的類別利用 line bot 顯示給使用者

✓ 將得出的類別
結合資料集去
查詢對應的食
譜並傳給使用
者



RESULT

根據使用者提供的食材
我們可以提供食譜給他們使用

文字 -> 食譜



```
if type=='text':
    msg = json_data['events'][0]['message']['text'] # 取得 LINE 收到的文字訊息
    recipy=pd.read_csv("data.csv")
    print(msg) # 印出內容

    temp=(recipy['Search']==msg)
    temp2=recipy.loc[temp]
    feback_title="title="+str(temp2['title'].head(1))
    feback_ingredients="ingredients="+str(temp2['ingredients'].head(1))
    feback_directions="directions="+str(temp2['directions'].head(1))
    reply_text=[TextSendMessage(text=feback_title+feback_ingredients+feback_directions)]
    line_bot_api.reply_message(tk,reply_text)#回傳訊息
```

圖片 -> 類別、食譜



感謝您的訊息！

很抱歉，本帳號無法個別回覆用戶的訊息。
敬請期待我們下次發送的内容喔

['cabbage', 'tomato']

title=6 Fried Cabbage
Name: title, dtype:
objectingredients=6 ["1 head cabbage", "1 onion", "1 bell pepper",...
Name: ingredients, dtype:
objectdirections=6 ["Chop or cut up in small pieces; let stand at...
Name: directions, dtype: object



感謝您的訊息！

很抱歉，本帳號無法個別回覆用戶的訊息。
敬請期待我們下次發送的内容喔

['egg']

title=38 Pecan Kisses
Name: title, dtype:
objectingredients=38 ["1 egg white", "2/3 c. light brown sugar", "1...
Name: ingredients, dtype:
objectdirections=38 ["Beat egg white until stiff.", "Add sugar and...
Name: directions, dtype: object

cabbage+tomato

感謝您的訊息！

很抱歉，本帳號無法個別回覆用戶的訊息。
敬請期待我們下次發送的内容喔

title=6 Fried Cabbage
Name: title, dtype:
objectingredients=6 ["1 head cabbage", "1 onion", "1 bell pepper",...
Name: ingredients, dtype:
objectdirections=6 ["Chop or cut up in small pieces; let stand at...
Name: directions, dtype: object

shrimp

感謝您的訊息！

很抱歉，本帳號無法個別回覆用戶的訊息。
敬請期待我們下次發送的内容喔

title=40 Shrimp Dip
Name: title, dtype:
objectingredients=40 ["8 oz. cream cheese", "1/2 c. mayonnaise", "1...
Name: ingredients, dtype:
objectdirections=40 ["Cream together cheese and mayo.", "Add shrim...
Name: directions, dtype: object

圖片 -> 類別、食譜

```
elif type=='image':

    msgID=json_data['events'][0]['message']['id'] #取得訊息id
    message_content=line_bot_api.get_message_content(msgID) #根據訊息ID取得訊息內容
    file_path=f'static/{msgID}.jpg'
    with open(file_path,'wb') as fd:#在同樣的資料夾中建立以訊息ID為檔名的.jpg檔案
        fd.write(message_content.content) #以二進位的方式寫入檔案
    img=cv2.imread(file_path)
    public_url='https://0bcb-122-100-89-165.ngrok-free.app/'+file_path #取得影片公開URL
    model = YOLO(r"C:\Users\clair\OneDrive\Desktop\李佩臻_期中報告\runs\detect\train\weights\best.pt")
    results = model (img, save=False, show=False, hide_labels=True)
    total_type=[]
    for i in results:
        if len(i.bboxes.xyxy)==0:
            answer="好像沒東西..."
            break;
        for j in i:
            start_point=(int(j.bboxes.xyxy.cpu().numpy()[0][0]),int(j.bboxes.xyxy.cpu().numpy()[0][1]))
            end_point=(int(j.bboxes.xyxy.cpu().numpy()[0][2]), int(j.bboxes.xyxy.cpu().numpy()[0][3]))
            if int(j.bboxes.cls.cpu().numpy()[0])==0:
                total_type.append("tomato")

            if int(j.bboxes.cls.cpu().numpy()[0])==1:
                total_type.append("egg")

            if int(j.bboxes.cls.cpu().numpy()[0])==2:
                total_type.append("shrimp")

            if int(j.bboxes.cls.cpu().numpy()[0])==3:
                total_type.append("cauliflower")

            if int(j.bboxes.cls.cpu().numpy()[0])==4:
                total_type.append("onion")
```

```
        if int(j.bboxes.cls.cpu().numpy()[0])==4:
            total_type.append("onion")

        if int(j.bboxes.cls.cpu().numpy()[0])==5:
            total_type.append("cabbage")

    total_type=list(set(total_type))
    answer=total_type
    if len(answer)<=1:
        R=answer[0]
    else:
        R=answer[0]+'+'
        for r in range(len(answer)-1):
            R+=total_type[r+1]
    recipy=pd.read_csv("data.csv")
    temp3=(recipy['Search']==str(R))
    temp4=recipy.loc[temp3]
    feback_title="title="+str(temp4['title'].head(1))
    feback_ingredients="ingredients="+str(temp4['ingredients'].head(1))
    feback_directions="directions="+str(temp4['directions'].head(1))
    feback=feback_title+feback_ingredients+feback_directions
    reply=[ #回傳影片和文字訊息
        TextSendMessage(text=str(answer)),
        TextSendMessage(text=str(feback))
        #設定要回傳的訊息
    ]

    line_bot_api.reply_message(tk,reply) #使用回復列表
except:
    print(body) # 如果發生錯誤，印出收到的內容
    return 'OK' # 驗證 Webhook 使用，不能省略

if __name__ == "__main__":
    app.run()
```